University of Rochester
Warner School of Education

Designing a Minecraft World That Carries Identity and Community Knowledge

Mini-Unit Packet

Layered Genius: Designing for Joy, Identity, Skills, and Intellect

Grade 4–5 Computer Science / Digital Literacy mini-unit centered on Minecraft Education, multimodal meaning-making, identity-safe design, debugging, disciplinary writing, and intellectual analysis

Student	Piter Zacari Garcia Bautista
Course	EDU498: Literacy Learning as Social Practice
Session Focus	Toward the Pursuit of Intellect
Assignment	CHRE Mini-Unit Extension
Mini-Unit Focus	Minecraft world-building as identity, community, coding literacy, disciplinary writing, and intellectual design analysis
Student Profile	Ethan, Student 5, Computer Science
Submission Date	June 2026

This revision extends the earlier mini-unit through the pursuit of Intellect. It treats Minecraft Education as a disciplinary literacy space where students build, code, debug, document, explain, and analyze how digital design choices communicate meaning to an audience.

Revision note: *New additions are shown in blue. This version keeps Joy, Identity, and Skills developed while strengthening Intellect through analysis, evaluation, reflection, and revision. Criticality remains in a brainstorm phase, as required for the current stage of the assignment.*

Mini-Unit Snapshot

Task: extend the mini-unit to include the pursuit of Intellect by asking students to analyze how digital design choices communicate identity, community knowledge, belonging, access, and audience meaning.

Grade / discipline: Grade 4–5 Computer Science / Digital Literacy

Student profile: Ethan, a Seneca / Haudenosaunee and mixed-identity student who enjoys coding, game design, robotics, collaboration, outdoor learning, and community storytelling.

Mini-unit focus: Students use Minecraft Education to plan and begin building a small digital place that communicates identity, community knowledge, belonging, or future possibility.

Essential question: How can code, design, and digital place-making communicate who we are, what matters to us, and what our communities know?

End goal/backward design target: By the end of the unit, students will share a Minecraft place and a short Developer Design Note explaining how specific build/code choices communicate meaning for an audience. Evidence includes the world/screenshot, plan, debug log, peer feedback, revision, and written/oral explanation.

Unit context and purpose: This three-lesson unit is designed for a Grade 4–5 Computer Science / Digital Literacy setting using Minecraft Education. Students use digital world-building as a literacy practice: they plan, code, debug, label, explain, analyze, and share a digital place that carries meaning. The purpose is to help students see coding as authorship, identity, community knowledge, audience-aware design, and intellectual meaning-making. The unit foregrounds Joy and Identity by giving students choice, visible progress, and a protected way to connect design with who they are or what they value. It strengthens Skills by making disciplinary language and writing explicit. This revision develops Intellect by asking students to analyze how code, signs, paths, colors, structures, repeated patterns, accessibility supports, and audience choices shape how others read a digital world. Criticality remains in a brainstorming phase, but the plan now names initial justice questions students will use to think about representation and access.

Three-lesson Arc:

- **Lesson 1: Plan a Meaningful Place.** Students compare a generic build with a meaningful digital place, choose a place concept, sketch meaningful features, decompose the build into steps, and draft a design goal that names the audience and purpose.
- **Lesson 2: Build, Code, Debug, and Analyze.** Students use Minecraft / MakeCode to build, test, revise, and document one coded or repeated feature that helps the world communicate meaning.
- **Lesson 3: Share, Explain, and Reflect.** Students present the world, submit a Developer Design Note, explain design choices, receive feedback, revise one feature for clarity, welcome, or accessibility, and reflect on representation.

Across the arc, the disciplinary thread stays consistent: students move from idea to plan, from plan to build/code, from code to debugging evidence, and from debugging evidence to an intellectual explanation of how design communicates meaning. This makes computer science skill visible without separating it from identity, joy, access, and community meaning.

Source Trail and Rationale for the Pick

This work grows from my work toward "*Layered Genius: Designing for Joy and Identity*" and from Muhammad's Identity pursuit, which frames identity as self-definition. It also carries forward my joy commitment of "belonging before building" and my placement observations of debugging, collaborative work, peer teaching, and student-led problem framing. I also draw on Moje, Luke, Davies, and Street's literacy-and-identity lens, Skerrett's attention to repositioning students as capable contributors, and Moje's disciplinary literacy framework. These sources help name coding, debugging, design, and explanation as literacy practices. Minecraft Education/MakeCode and the CSTA/ISTE standards serve as practical anchors for sequencing, debugging, computational thinking, creative communication, collaboration, and multimodal digital design.

For this revision, I also use the Class 9 pursuit of Intellect to move the mini-unit beyond building and documentation. Students are not only asked to create a Minecraft world; they are asked to analyze how digital choices carry meaning for an audience, evaluate whether the world communicates clearly and respectfully, and refine one feature based on feedback. This keeps the task rigorous while making the rigor meaningful and reachable.

Student Profile and Identity Safety

Ethan's profile matters because the lesson can recognize him as a coder, designer, storyteller, and community-minded problem solver. At the same time, the lesson should honor his identity without requiring him to explain, display, or represent his culture for the class. Instead, it offers identity-safe choice by allowing students to build from lived experience, imagination, community, interests, family, land, language, hobbies, or future hopes. Because Ethan's profile includes Seneca / Haudenosaunee heritage, culturally specific symbols or stories should be student-chosen and handled with care. Private disclosure is never required. Fictionalized, symbolic, or aspirational places are valid.

Identity design move: The lesson treats identity as a design resource, not a disclosure requirement. Students choose what counts as meaningful, and the teacher assesses how clearly design choices communicate meaning. This keeps the work aligned with our focus on identity while respecting privacy, agency, and student control.

Discipline Standards, Revised Pursuit, and Learning Activity Ideas

Learning standard CSTA Computer Science:

1B-AP-17: Describe choices made during program development using code comments, presentations, and demonstrations.

Revised: Students will design, evaluate, and refine a Minecraft Education world that communicates identity, community knowledge, belonging, or future possibility. Students will analyze how their code, signs, paths, structures, repeated patterns, and audience supports shape how others understand the digital space, and they will explain their design choices through a short presentation, demonstration, or Developer Design Note.

Digital Representation and Access to Education: Addressing how digital spaces can include or exclude students' identities, community knowledge, and access needs while promoting respectful and accessible design. Example: Creating a Minecraft Education world that represents a meaningful place, community story, or future possibility without requiring students to disclose private cultural or personal information.

Disciplinary literacy move: In this mini-unit, writing is not separate from computer science. Students write like beginning designers/developers: they state a design goal, name steps, describe a bug or revision, explain how a feature works, and connect technical choices to audience meaning.

Pursuits Alignment

Pursuit	How to Address it
Joy	Students experience choice, visible progress, peer support, debugging as discovery, and the satisfaction of building something shareable.
Identity	Students design a meaningful place and explain how specific features communicate belonging, community, memory, culture, interests, land, imagination, or future possibility with dignity.
Skills	Students practice sequencing, decomposition, repeated patterns, debugging, reasoning, screenshot use, partner communication, and oral/written explanation. They also complete a Developer Design Note that documents their design goal, steps, bug/revision, and audience meaning.
Intellect	Students analyze how code, space, signs, paths, color, structures, repeated patterns, interaction, accessibility supports, and audience choices work together to create meaning in a digital world. They evaluate whether their world communicates clearly, revise one feature based on feedback, and explain how the design helps an audience understand identity, community knowledge, belonging, or future possibility.
Criticality	Brainstorm phase: students ask whose stories appear in digital worlds, whose knowledge is missing, how accessibility shapes belonging, and what respectful use means when cultural symbols or community knowledge are involved. Criticality is named here as a future extension rather than fully developed in this lesson sequence.

Multimodal and Multiliteracies Design

Minecraft functions here as a text students author. They read digital space, sketch a plan, build visually, use procedural thinking, explain orally, label in writing, analyze design choices, and revise through peer feedback.

Modes included: spatial design, visual design, coding/sequencing, oral explanation, written labels or reflection, screenshots, collaboration, peer feedback, and audience-aware revision.

Access moves: sketch-first or build-first choice; projected model; partner roles; sentence stems; optional oral explanation; screenshots as memory support; chunked directions; low-floor/high-ceiling challenge options; and a small revision target instead of requiring students to rebuild the whole world.

Why this supports multiliteracies: Students are not only writing or answering questions; they are composing meaning across space, image, code, talk, labels, screenshots, and peer feedback. A path, sign, repeated pattern, gathering space, accessibility feature, or coded sequence can all function as evidence of meaning-making.

Intellectual literacy layer: Students read their world as designers by asking what an audience can infer from paths, signs, repeated patterns, coded movement, points of interaction, and accessibility supports. They write as beginning developers by documenting design goals, build/code steps, bugs, revisions, and audience meaning. The multimodal product and the written note work together: the world shows the design, while the note explains the technical, cultural, and intellectual choices behind it.

Lesson 1 Plan

Duration: 60 minutes

Lesson title: Planning and Prototyping a Meaningful Minecraft Place

Student-friendly target: I can plan and begin building a Minecraft place that shows something important about identity or community, and I can explain how my design choices carry meaning.

Identity-safe teacher language: Your place can be real, imagined, symbolic, collaborative, or future-facing. You do not have to share anything private. The goal is to make thoughtful design choices and explain what those choices communicate.

Intellect bridge: Students compare a generic Minecraft structure with a meaningful digital place and ask what makes a digital space readable, welcoming, accessible, or meaningful to an audience. This frames building as intellectual design.

Objectives:

1. Students will choose a meaningful place connected to identity, community, belonging, or future possibility.
2. Students will break the place into smaller build/code steps.
3. Students will analyze how a sign, path, structure, color, repeated pattern, or coded feature can guide an audience's understanding.
4. Students will draft the first part of a Developer Design Note by naming their design goal and first 3–5 build/code steps.

Mins	Activity	Purpose
0–7	Compare a generic Minecraft build with a meaningful digital place. Ask what makes a place meaningful, welcoming, or accessible.	Frames design as meaning-making and analysis.
7–15	Model a small place plan with entrance, sign, path, gathering spot, and repeated pattern. Think aloud about what each feature communicates.	Makes audience and design choices visible.
15–38	Students choose a real, imagined, symbolic, or future place; sketch 3–5 features; and label first build/code steps.	Builds identity-safe agency and decomposition.
38–50	Students begin a prototype, test one feature, and add one note explaining what the feature helps the audience understand.	Connects design to audience meaning.
50–60	Partner share and exit slip: "This helps my audience understand..." and "My next build/code step is..."	Captures oral and written evidence.

Differentiation: Students may work alone or with roles such as builder, navigator, debugger, or explainer. Beginners may build one strong scene; advanced students may add loops, agent movement, signage, or an audience-centered accessibility feature.

Lesson 2 Plan

Duration: 60 minutes

Lesson title: Building, Coding, Debugging, and Analyzing the Meaningful Place

Student-friendly target: I can build and revise my Minecraft place so that at least one coded or repeated feature helps communicate meaning.

Joy + identity bridge: Lesson 2 keeps coding joyful by treating debugging as experimentation rather than failure. Students revise features so their worlds become clearer, more welcoming, or more meaningful to an audience.

Intellect bridge: Students think like designers and beginning developers. They do not only ask, "Does it work?" They also ask, "What does this design choice help my audience understand?"

Objectives:

1. Students will build at least two meaningful features from their plan.
2. Students will use a sequence, loop, repeated structure, or agent-based action to create or improve the world.
3. Students will debug one issue by naming the problem, testing a change, and recording what changed.
4. Students will analyze how one code/build choice affects audience understanding, welcome, or access.

Mins	Activity	Purpose
0–8	Students choose one must-build feature and one try-to-code feature, then name what the feature should help the audience understand.	Connects coding/design to meaning.
8–15	Teacher models a debug cycle and how to write what the revision communicates.	Makes debugging visible, writable, and purposeful.
15–42	Students build and code with partner roles. Teacher asks: "What does this choice help your audience read or understand?"	Develops skill and analysis.
42–52	Screenshot checkpoint: students label one feature, explain how it works, and explain what it communicates.	Connects product to intellectual explanation.
52–60	Quick share: one success, one bug, one next step, and one audience meaning.	Builds design language and community expertise.

Disciplinary writing evidence: Students complete the middle section of the Developer Design Note: "My bug or design problem was..." "I expected..." "What happened was..." "I changed..." "This helped my audience understand..."

Lesson 3 Plan

Duration: 60 minutes

Lesson title: Sharing Worlds, Reading Design, and Reflecting on Representation

Student-friendly target: I can present my Minecraft place, explain how it carries identity or community knowledge, and respond respectfully to other students' designs.

Representation bridge: Before students share, the teacher names that digital worlds are designed from choices. Some choices make people feel welcomed and seen; other choices erase people, flatten culture, or make access harder. Students explain what they chose, not private identity.

Objectives:

1. Students will explain at least two design choices and how they carry meaning.
2. Students will give peer feedback using notice/wonder language.
3. Students will analyze how digital worlds can include, exclude, or misrepresent people and communities.
4. Students will revise one feature for clarity, welcome, or accessibility.

Mins	Activity	Purpose
0–8	Students prepare a two-sentence explanation and choose one screenshot or live view to show.	Supports concise presentation.
8–35	Gallery walk or paired screen-share. Students explain their worlds and collect one notice/wonder comment.	Centers oral, visual, and digital literacy.
35–48	Whole-group brainstorm: "Whose stories do digital worlds usually show, whose could be missing, and how can designers make spaces more respectful and accessible?"	Keeps Criticality in brainstorm phase while making the representation visible.
48–56	Students revise one sentence, sign, pathway, or accessibility feature based on feedback and explain why the change matters.	Turns feedback into intellectual action.
56–60	Exit reflection: "My world shows..." "A designer can make digital spaces more welcoming by..." "One feature I revised was..."	Captures final evidence.

Student reflection and analysis prompts:

- What does your Minecraft world help an audience understand?
- Which design choice carries the most meaning, and why?
- How do your paths, signs, colors, structures, repeated patterns, or code guide the audience?
- How can a digital world make someone feel welcomed or excluded?
- Whose stories are easy to represent in digital worlds, and whose stories are often missing?
- How can designers avoid using culture, identity, or community knowledge as decoration?
- What did you revise after feedback, and how did that change your world's meaning?

Developer Design Note prompt: "My design goal was..." "The build/code steps I used were..." "One bug or revision I worked through was..." "One feature I want my audience to notice is..." "This world communicates identity, community, belonging, or future possibility by..." "One revision I made for clarity, welcome, or access was..."

Assessment and Evidence

Evidence collected across the unit: planning sheet, debug log, screenshots or sketches, teacher conference notes, peer feedback, short presentation, Developer Design Note, and exit reflection.

Criterion	**Meeting**	**Developing**
Meaningful design	World includes features connected to identity, community, belonging, or future possibility.	World has features, but the meaning needs clearer explanation.
Computational thinking	Student decomposes, sequences, tests, debugs, or uses a repeated/coded feature.	Student builds but needs support naming steps or debugging.
Disciplinary writing	Student documents design goal, steps, bug/revision, and audience meaning using beginning computer science language.	Student explains the project generally but needs support connecting technical steps to meaning.
Intellectual analysis	Student analyzes how design choices such as paths, signs, structures, colors, repeated patterns, code, or accessibility supports help an audience read the world.	Student names a design choice but needs support explaining how it affects audience understanding.
Critical awareness	Student can brainstorm whose stories are represented, whose stories may be missing, and how design can welcome an audience.	Student notices audience or representation with prompting.

How the evidence is read: The teacher does not grade cultural disclosure or artistic polish. The main evidence is whether students connect a design choice to a purpose, use basic computer science language to describe process, analyze how choices affect audience meaning, and revise after testing or feedback.

Sample and Appendix Tools

Teacher model sample: A small community garden world with an entrance path, welcome sign, gathering circle, repeated fence pattern, and one coded/agent-assisted repeated feature. The model should be meaningful but not perfect so students see revision as normal.

Student-facing routine: Students use one tool at a time: plan first, debug next, then complete the Developer Design Note and share card. This keeps the workflow accessible while preserving a full evidence trail.

Reflection

This mini-unit shows how digital literacy can become a space for belonging. Joy is built into choice, creation, debugging, peer support, and visible progress. Identity is built into meaningful place-making and student control over what is shared. Skills are built through sequencing, decomposition, debugging, documentation, and presentation. Intellect is built when students analyze how those technical and design choices communicate meaning to an audience.

The revision strengthens the mini-unit by making intellectual analysis visible. Students are not only building in Minecraft; they are learning how designers and beginning developers communicate a design goal, document a process, test and debug, respond to feedback, and explain how technical choices affect an audience. They learn that code is not only functional. Code, space, signs, paths, colors, structures, repeated patterns, and accessibility supports can all shape how a digital world is read.

For Ethan specifically, the design matters because it does not ask him to simplify himself into one identity label. It gives him a way to connect coding, game design, land, community, mixed identity, and cultural knowledge only to the extent he chooses. That makes the technical work more humane: code becomes a tool for meaning-making, not a gatekeeping measure.

For my own teaching, the most important design decision is that students are assessed on how intentionally they communicate meaning. That distinction matters for neurodivergent, culturally connected, multilingual, and mixed-identity students because it keeps literacy tied to agency. In this unit, joy is not separate from rigor: students still decompose, debug, revise, document, analyze, and explain, but those skills serve a world they care about.

This pursuit expands students' intellectualism because it helps them understand that design is a form of thinking. Students learn to read digital spaces as texts and to understand that technical choices are not neutral. A design can welcome, confuse, include, exclude, preserve, flatten, or erase meaning. The work prepares students to see themselves not just as users of technology, but as designers whose choices affect others.

Appendix: Working-Memory-Friendly Lesson Flow Figures

Design note: These visual figures are meant to support teachers and students who benefit from seeing the whole learning path at once. Each figure separates the build sequence, teacher support, and evidence so the lesson can be followed without holding every step in working memory.

Notice meaning → Choose place → Sketch features → Begin prototype → Explain choice

Teacher support: Model decomposition, protect private identity choices, offer roles, and show the first build step.

Evidence: Planning card, prototype or sketch, partner explanation, and exit slip.

Figure 1: Lesson 1 visual flow: Plan a Meaningful Place.

Pick feature → Build and code → Test world → Debug issue → Label change

Teacher support: Normalize debugging, conference with short prompts, and connect technical changes back to meaning.

Evidence: Debug log, before/after note, screenshot label, and oral explanation.

Figure 2: Lesson 2 visual flow: Build, Code, and Debug.

Prepare share → Present world → Collect feedback → Revise for welcome → Reflect on design

Teacher support: Provide sentence stems, protect choice, guide honest feedback, and invite students to revise a small piece.

Evidence: Final screenshot or live view, peer feedback note, revision note, and exit reflection.

Figure 3: Lesson 3 visual flow: Share, Revise, and Reflect.